

## Full Length Article

# Foreground guided dual branch learning for sparse 4D radar 3D object detection

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## ABSTRACT

4D imaging radar offers long-range perception and Doppler velocity measurement capabilities, while maintaining strong robustness in nighttime scenes and adverse weather conditions such as rain and fog. These advantages make it a promising sensor for 3D perception in autonomous driving. However, due to radar imaging mechanisms, 4D radar point clouds are usually highly sparse and unevenly distributed, and are often contaminated by background clutter and isolated noise points. As a result, real objects are represented by only a few valid points and can be easily overwhelmed by complex background responses. Existing methods mainly focus on global feature enhancement or local denoising, while paying insufficient attention to explicit modeling of point-wise foreground characteristics. To address these issues, this paper proposes a foreground guided dual branch learning framework for 3D object detection in sparse 4D radar scenarios. Specifically, a differentiable learning-based query generator jointly models local density and motion consistency to estimate point-wise foreground importance and generate foreground aware cues. Then, a foreground guided gated cross branch fusion mechanism selectively injects auxiliary information into potential object regions, thereby enhancing object responses and suppressing background interference. Finally, at the proposal stage, cylindrical geometric constraints and a foreground prioritized keypoint selection strategy are combined to improve local feature representations. Experimental results on VoD and TJ4DRadSet demonstrate that the proposed radar-only method achieves competitive performance, even comparable to several radar-camera fusion methods, while maintaining a favorable balance between detection accuracy and inference efficiency.

## 1. Introduction

3D object detection is a core task in autonomous driving perception, and its results directly affect subsequent path planning and safety critical decision making. Compared with optical sensors such as cameras, millimeter wave radar is more robust to illumination changes and adverse weather conditions, and is well suited for long range sensing, making it highly valuable in complex traffic environments (Dong et al., 2026; Li et al., 2023; 2025b; Wang et al., 2026). In recent years, 4D imaging radar has extended conventional radar by incorporating elevation information and radial velocity measurements. It can perceive both the 3D spatial locations and motion states of objects, thereby providing richer perceptual cues for dynamic scene understanding (Liu et al., 2023c).

Compared with conventional LiDAR, 4D radar provides complementary advantages for autonomous driving perception. As shown in Fig. 1, it provides Doppler velocity measurements for motion aware perception, supports long range sensing, and is less sensitive to adverse illumi-

nation and weather conditions. Owing to these characteristics, 4D radar has become an important research direction in autonomous driving 3D perception (Lin et al., 2024; Zheng et al., 2023).

Nevertheless, directly applying 4D radar to 3D object detection remains challenging. Due to the inherent limitations of radar imaging mechanisms, 4D radar point clouds are usually highly sparse and unevenly distributed, and are often accompanied by background clutter and isolated noise points (Bi et al., 2025a; Zhong et al., 2025). This issue becomes more severe in long range and small object scenarios, where objects may contain only a few valid points, while background regions may contain numerous irrelevant reflections. As a result, detection models need to recover discriminative representations from extremely limited object points while suppressing interference from complex backgrounds (Musiat et al., 2024).

Existing studies mainly focus on global feature enhancement, local denoising, or proposal refinement, while paying relatively insufficient attention to explicit modeling of point-wise foreground characteristics

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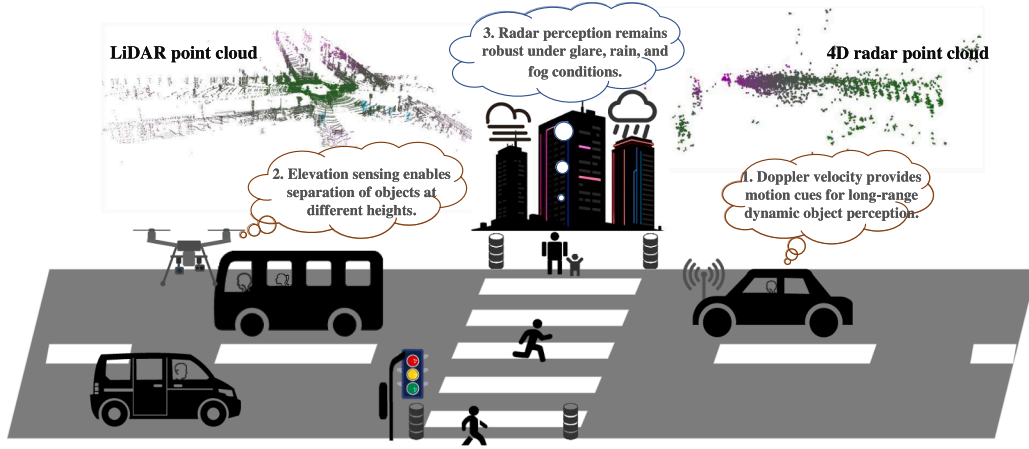


Fig. 1. Illustration of 4D radar perception advantages in complex traffic scenarios.

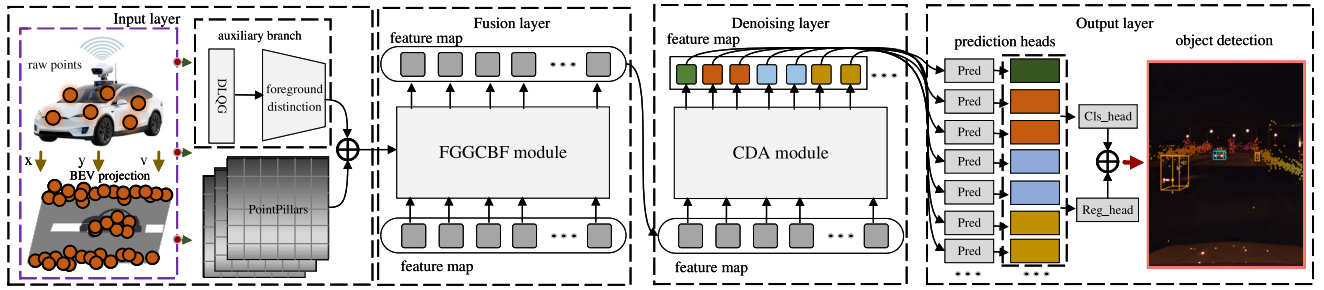


Fig. 2. Overall architecture of the proposed foreground guided dual branch learning framework.

(Chen et al., 2023; Li et al., 2025a). Most existing methods encode radar point clouds as a unified whole and lack an early stage mechanism for distinguishing foreground and background points. Consequently, object related responses are difficult to highlight during feature construction, especially in sparse and cluttered radar scenes.

Foreground modeling has been widely explored in fields such as image segmentation, where object related regions are explicitly distinguished from irrelevant backgrounds to improve feature discrimination (Lei et al., 2025; Zhang et al., 2025). For 4D radar point clouds, the imbalance between weak foreground responses and complex background clutter naturally motivates explicit foreground modeling. By identifying high value points in advance, irrelevant interference can be suppressed more effectively (Zhang et al., 2024).

To address the above issues, this paper introduces a foreground background distinction mechanism into 4D radar 3D object detection and proposes a foreground guided dual branch learning framework for sparse radar scenarios. The proposed method mines point-wise foreground cues from raw radar point clouds and uses them to guide cross branch feature fusion and proposal level keypoint refinement. In this way, the proposed framework forms a unified optimization pipeline from point level foreground selection, to feature level selective fusion, and then to proposal level geometric constraints.

The main contributions are summarized as follows:

- We propose a foreground guided dual branch learning framework for 4D radar 3D object detection. By introducing explicit foreground background distinction into the detection pipeline, the proposed framework improves object representation in sparse radar scenarios.
- We design a differentiable learning-based query generator and a foreground guided gated cross branch fusion module. The former generates foreground cues by modeling local density and motion consistency, while the latter guides cross branch feature interaction to focus on potential object regions.

- At the proposal stage, we introduce a foreground prioritized key-point selection strategy based on cylindrical geometric constraints. This strategy enhances the representation of object related regions and reduces interference from background clutter and noise points.
- Experiments on VoD and TJ4DRadSet show that the proposed radar only method achieves competitive accuracy against radar-based and several radar-camera fusion methods while retaining real-time efficiency.

## 2. Related work

### 2.1. Camera-based 3D object detection

Camera-based 3D object detection is an important research direction in autonomous driving perception. Since cameras provide rich texture and semantic information, early methods attempted to estimate 3D bounding boxes from monocular images (Chen et al., 2016). With the development of multi-view perception, research has gradually shifted from explicit geometry-based methods to bird's eye view (BEV) representation-based approaches. For example, LSS (Phillion & Fidler, 2020) projects multi-view image features into the BEV space, providing an effective paradigm for vision-based BEV detection. DETR3D (Wang et al., 2022) and PETR (Liu et al., 2022) further introduce query mechanisms and cross view feature interactions to enhance object modeling in 3D space. SparseBEV (Liu et al., 2023a) employs a sparse query strategy to improve detection efficiency and model global relationships.

Overall, camera-based methods have clear advantages in semantic representation and global context modeling. However, images do not provide direct depth measurements and are easily affected by depth estimation errors, occlusion, illumination changes, and scale ambiguity. Therefore, in complex traffic scenarios, relying solely on image infor-

mation makes it difficult to stably recover accurate 3D geometric structures.

## 2.2. LiDAR-based 3D object detection

Compared with images, LiDAR can directly capture the 3D geometric information of objects and surrounding scenes. It has therefore long been one of the mainstream sensors for 3D object detection in autonomous driving (Valverde et al., 2025). To efficiently represent irregular point clouds, existing studies have developed several technical routes, including point-based, voxel-based, and point voxel hybrid methods. The PointNet series (Qi et al., 2017) directly learns features from raw point sets, laying the foundation for point level feature extraction. VoxelNet (Zhou & Tuzel, 2018) and SECOND (Yan et al., 2018) convert unordered point clouds into regular 3D grids through voxelization, enabling the use of 3D convolutions for point cloud detection. PointPillars (Lang et al., 2019) further adopts a pillar-based representation to compress 3D point clouds into the BEV space, significantly improving inference efficiency while maintaining detection accuracy. PV-RCNN (Shi et al., 2020) combines keypoint feature aggregation with proposal refinement, further improving the localization accuracy of 3D bounding boxes.

These studies have established mature techniques for spatial geometric modeling and BEV representation construction, and their designs also provide an important foundation for 4D radar point cloud detection. However, LiDAR performance can degrade under adverse environmental conditions such as rain, fog, snow, and dust. In addition, its relatively high cost limits its large scale deployment in practical autonomous driving systems.

## 2.3. 4D radar-based 3D object detection

4D radar has recently attracted increasing attention in autonomous driving due to its robustness under adverse illumination and weather conditions, as well as its ability to provide elevation and radial velocity measurements (Yang et al., 2026). Nevertheless, 4D radar point clouds are still typically sparse, unevenly distributed, and easily contaminated by multipath reflections and background clutter. These limitations become more pronounced in long range and small object scenarios, where object related responses may be overwhelmed by irrelevant reflections, posing challenges for robust 3D object detection (Wu et al., 2026).

To address these issues, existing studies have improved 4D radar detection performance from different perspectives. Early methods usually adapted point cloud detection frameworks such as SECOND (Yan et al., 2018) and PointPillars (Lang et al., 2019) to radar point clouds, demonstrating the feasibility of voxel-based and pillar-based representations for 4D radar detection. Subsequently, RadarPillarNet (Musiat et al., 2024) enhanced dynamic object representations by modeling the spatial and velocity attributes of radar points. SMURF (Liu et al., 2023b) employed multi-scale representations and density modeling to alleviate feature degradation caused by point cloud sparsity and uneven spatial distribution. DADAN (Wang et al., 2025) introduced dynamic augmentation and density aware mechanisms to improve the discriminative capability for moving objects. Radar M3-Net (Yang et al., 2025) further incorporated multi-frame information for temporal modeling, leveraging the motion sensing advantage of radar. MAFF-Net (Bi et al., 2025a) enhanced object representations in sparse radar point clouds through auxiliary feature fusion and geometric constrained refinement.

Although these methods have made effective progress in feature enhancement, density modeling, temporal fusion, and region level denoising, most of them still improve radar representations mainly at the global feature or proposal level, leaving explicit point-wise foreground modeling insufficiently explored. DADAN (Wang et al., 2025) attempted to enhance foreground representations using dynamic information, but its focus remains on local feature encoding and it does not explicitly use foreground cues to guide subsequent feature fusion. For 4D radar

point clouds, objects are often represented by only a small number of points, while background clutter is abundant. Without early distinction between foreground and background points, the network is easily disturbed by irrelevant reflections during feature construction. Motivated by this observation, the proposed method estimates point-wise foreground importance and uses foreground cues to guide cross branch fusion and keypoint selection, thereby enhancing object related responses and suppressing background interference.

## 3. Method

### 3.1. Overview

To address the challenges of sparse target points and background clutter interference in 4D radar point cloud detection, we propose a foreground guided dual branch learning framework (FGDBL). As illustrated in Fig. 2, the proposed framework consists of five main components: a main branch radar point cloud representation module, an auxiliary branch differentiable learning-based query generator (DLQG), a foreground guided gated cross branch fusion module (FGGCBF), a cylindrical denoising auxiliary module (CDA), and a detection head. The proposed modules are implemented within a PV-RCNN-style two stage detection framework (Shi et al., 2020), where the first stage generates BEV features and 3D proposals, and the second stage refines proposal level representations.

The overall pipeline is summarized as follows. First, the raw point cloud is divided into pillar units, and a main branch base representation is generated through pillar feature encoding and BEV mapping. Then, in the auxiliary branch, DLQG extracts potential foreground cues from sparse radar point clouds. Guided by these cues, FGGCBF selectively injects foreground information from the auxiliary branch into the main branch, thereby enhancing responses in object regions while suppressing background interference. Next, CDA performs keypoint sampling and multi-source feature aggregation within proposal regions. Finally, the detection head predicts the categories and 3D bounding box parameters of objects.

### 3.2. Main branch radar point cloud representation

#### 3.2.1. Pillar feature encoding for radar point clouds

Let the input 4D radar point cloud be denoted as:

$$\mathcal{P} = \{p_i\}_{i=1}^N, \quad p_i = (x_i, y_i, z_i, v_i), \quad (1)$$

where  $N$  denotes the number of points,  $(x_i, y_i, z_i)$  represents the 3D coordinates of the  $i$ th point, and  $v_i$  denotes its radial velocity.

To enable efficient processing of sparse radar point clouds, we adopt a pillar partitioning strategy similar to PointPillars (Lang et al., 2019), which discretizes the point cloud into regular 2D grids in the BEV plane. Let  $\mathcal{V}_j$  denote the point set within the  $j$ th non-empty pillar. The pillar feature encoding process can then be formulated as:

$$\mathbf{h}_j = \phi_{\text{RP}}(\mathcal{V}_j), \quad (2)$$

where  $\phi_{\text{RP}}(\cdot)$  denotes the PointPillars style pillar feature encoding process that maps the radar points within each non-empty pillar to a feature representation, and  $\mathbf{h}_j$  is the high dimensional feature representation of the  $j$ th pillar.

Through feature encoding, the original irregular point set is transformed into a structured pillar representation, providing a unified input for subsequent BEV representation construction.

#### 3.2.2. BEV mapping and main branch feature construction

After obtaining the pillar features, we further project the non-empty pillars onto the BEV grid to construct the main branch BEV representation. This process can be formulated as:

$$\mathbf{F}_m = \phi_b(\{\mathbf{h}_j\}), \quad (3)$$

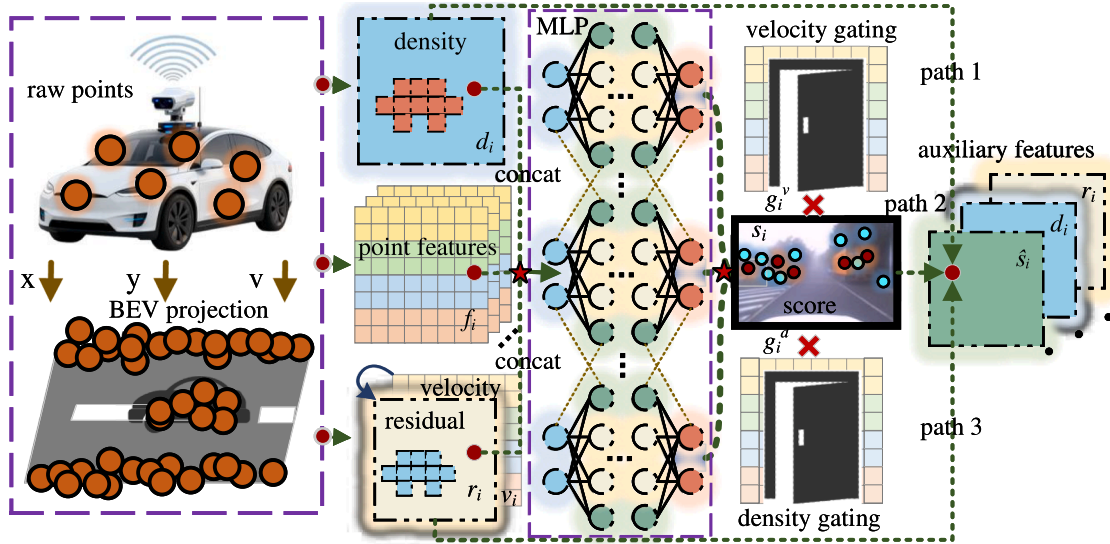


Fig. 3. Architecture of the differentiable learning-based query generator.

where  $\phi_b(\cdot)$  denotes the BEV mapping and spatial enhancement operation, which scatters the non-empty pillar features to their corresponding BEV grid locations and further enhances them through a 2D convolutional backbone, and  $\mathbf{F}_m \in \mathbb{R}^{C \times H \times W}$  is the output feature map of the main branch. Here,  $C$ ,  $H$ , and  $W$  denote the number of channels, height, and width, respectively.

The main branch aims to provide a stable spatial representation for the entire network. It should be noted that this stage mainly extracts general scene information from the raw radar point cloud, but does not explicitly distinguish foreground regions from background regions. Therefore, relying only on main branch features is still insufficient to fully address the challenges caused by sparse object points and background clutter in 4D radar point clouds. An auxiliary branch is further required to mine potential foreground cues.

### 3.3. Differentiable learning-based query generator

In 4D radar scenarios, real objects are usually represented by only a small number of points. This issue is particularly severe for long range objects and small objects such as pedestrians and cyclists, where the local point density is often very low. If a hard threshold-based query construction strategy is adopted, sparse foreground points may be mistakenly regarded as noise, thereby weakening the foreground guidance capability in the subsequent fusion stage. To address this problem, we design a differentiable learning-based query generator, as shown in Fig. 3. It jointly models local density and motion consistency to generate queries in a learnable manner.

Specifically, let  $\mathbf{f}_i$  denote the feature representation of the  $i$ th radar point, and let  $v_i$  denote its radial velocity. We first project the points onto the BEV grid and compute the local density and average velocity within the local neighborhood  $\mathcal{N}_i$ . Here,  $\mathcal{N}_i$  denotes the set of radar points contained in a  $5 \times 5$  BEV grid window centered on the grid cell containing the  $i$ th point. To reduce the influence of individual noisy points on velocity estimation, we adopt a neighborhood averaged velocity estimation strategy. The local motion state at each position is represented by the neighborhood average velocity:

$$\bar{v}_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} v_j, \quad d_i = |\mathcal{N}_i|. \quad (4)$$

Here,  $|\mathcal{N}_i|$  denotes the number of radar points contained in the local neighborhood  $\mathcal{N}_i$ ,  $\bar{v}_i$  smooths velocity perturbations caused by random noise within the local neighborhood, while  $d_i$  describes the structural density of the region where the point is located.

Considering that points belonging to the same object usually exhibit consistent motion properties, we further define the velocity residual of each point as:

$$r_i = |v_i - \bar{v}_i|. \quad (5)$$

A smaller  $r_i$  indicates that the point is more consistent with the local motion pattern and is therefore more likely to belong to a real object. In contrast, if the velocity distribution is disordered, the point is more likely to originate from noise or background clutter. Based on this observation, we feed the point feature, local density, and velocity residual into a lightweight scoring network to obtain the point importance score:

$$s_i = \sigma(\text{MLP}([\mathbf{f}_i, d_i, r_i])), \quad (6)$$

where  $\sigma(\cdot)$  denotes the sigmoid function.

To further enhance the physical interpretability of the scoring process, we introduce a velocity consistency gate and a density gate on top of the learned score. The final point-wise query score is defined as:

$$\hat{s}_i = s_i \cdot (\beta + (1 - \beta)g_i^v) \cdot (\beta + (1 - \beta)g_i^d), \quad (7)$$

where  $\beta$  is a lower bound coefficient that ensures a certain response is still retained in regions with low density or low consistency. The two gating terms are defined as:

$$g_i^v = \exp\left(-\frac{r_i}{\tau_v}\right), \quad g_i^d = \tanh\left(\frac{d_i}{\tau_d}\right). \quad (8)$$

Here,  $\tau_v$  and  $\tau_d$  are temperature coefficients controlling the gating responses. The velocity consistency gate  $g_i^v$  approaches 1 when the motion consistency is high, which helps suppress responses from velocity disordered points and highlight motion consistent candidates. The density gate  $g_i^d$  increases with local structural density, enhancing responses in structurally stable regions while suppressing isolated noisy points. The two gates adopt different functions to match their opposite monotonic relationships with point reliability: the exponential function decreases with the velocity residual, whereas the tanh function increases with local density and gradually saturates. The lower bounded gating design prevents low density but potentially valid foreground points from being completely suppressed. In this way, the query score preserves data driven learnability while explicitly incorporating motion and structural priors in 4D radar scenarios.

During query construction, we select query points according to  $\hat{s}_i$  using a Top-K strategy. The selected high score points provide foreground cues, while the main branch still preserves scene level contextual information from the original radar point cloud. This allows the model

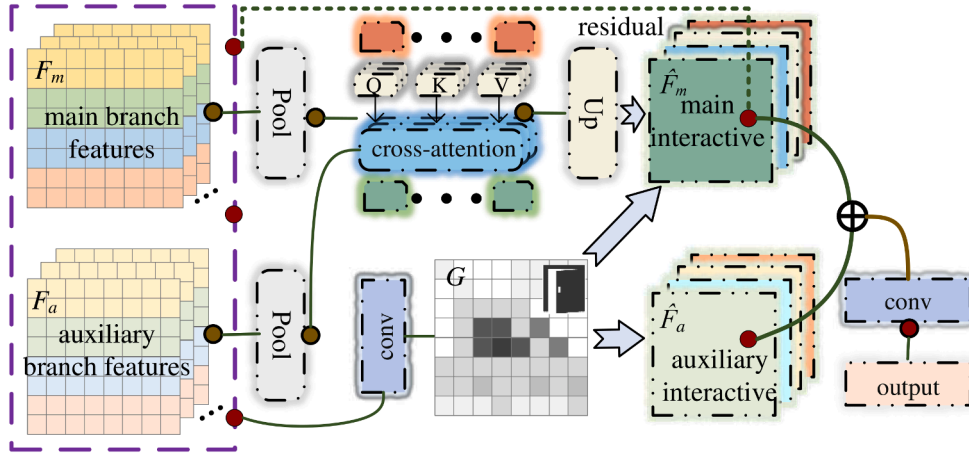


Fig. 4. Architecture of the foreground guided gated cross branch fusion module.

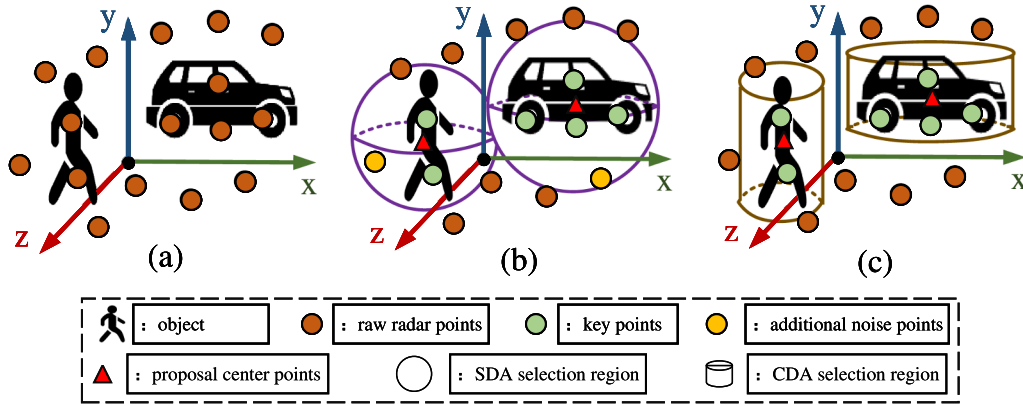


Fig. 5. Comparison of keypoint selection under spherical and cylindrical constraints.

to exploit object cues without losing global scene context. It should be noted that the point-wise scoring process in DLQG is differentiable, while Top-K selection is only used to construct foreground cues for subsequent modules. Meanwhile, the local average velocity, local density, and point scores are aggregated into a BEV query map, which is further encoded into the foreground query representation  $F_a$  for subsequent fusion. This design avoids the adverse effects of hard threshold filtering on sparse objects and helps stably preserve effective foreground cues in complex noisy environments.

### 3.4. Foreground guided gated cross branch fusion module

As described in Sections 3.2 and 3.3, we obtain the main branch feature and auxiliary branch feature, denoted as  $F_m, F_a \in \mathbb{R}^{C \times H \times W}$ . Here,  $F_m$  provides the geometric structure and scene context required for detection, while  $F_a$  provides stronger responses to potential object regions. To ensure that the auxiliary information contributes to object representation rather than introducing additional interference in background regions, we design a foreground guided gated cross branch fusion module, as shown in Fig. 4. The core idea is to enhance cross branch interaction only in potential object regions, while keeping the main branch features stable in background regions.

First, the features of the two branches are downsampled to a lower resolution and then fed into a cross branch attention interaction module. This establishes global correlations with a relatively low computational cost and produces a coarsely enhanced representation for the main branch:

$$F_c = \text{Up}(\text{Attn}(\text{Pool}(F_m), \text{Pool}(F_a))). \quad (9)$$

Here,  $\text{Pool}(\cdot)$  denotes the down sampling mapping,  $\text{Attn}(\cdot)$  denotes cross branch feature interaction, and  $\text{Up}(\cdot)$  restores the feature map to the original spatial resolution. This step mainly establishes coarse grained global correlations indicating where the network should focus.

Meanwhile, the auxiliary branch feature is used to generate a spatial gating map:

$$G = \sigma(\text{Conv}_{1 \times 1}(F_a)), \quad (10)$$

where  $G \in [0, 1]^{1 \times H \times W}$  represents the position dependent fusion strength, and  $\sigma(\cdot)$  denotes the sigmoid function. For potential object regions,  $G$  takes larger values, allowing more auxiliary information to be injected. For background regions,  $G$  takes smaller values, thereby suppressing the propagation of irrelevant information.

Based on the gating map, the main branch is updated as:

$$\hat{F}_m = F_m + G \otimes F_c, \quad (11)$$

while only the foreground related components are retained in the auxiliary branch:

$$\hat{F}_a = G \otimes F_a. \quad (12)$$

where  $\otimes$  denotes element wise multiplication. In this way, the fusion process no longer uniformly enhances the entire feature map. Instead, it selectively injects auxiliary information around potential object regions. Intuitively, this process uses an object region mask to constrain the fusion behavior, so that the interaction between the two branches mainly serves object representation while preventing background clutter from contaminating the main branch as much as possible.

Finally,  $\hat{\mathbf{F}}_m$  and  $\hat{\mathbf{F}}_a$  are concatenated and passed through a convolutional mapping layer to obtain the final fused feature:

$$\mathbf{F}_{\text{out}} = \text{Conv}_{3 \times 3}(\hat{\mathbf{F}}_m, \hat{\mathbf{F}}_a). \quad (13)$$

This design balances global information exchange and fine grained local control. The former improves the representation capability of potential object regions, while the latter reduces ineffective fusion in background regions. As a result, the negative effects of noise propagation and interference accumulation in sparse scenes are alleviated.

Afterward, following the RPN design used in PointPillars (Lang et al., 2019), we generate 3D proposals from the fused features for subsequent refinement.

### 3.5. Cylindrical denoising auxiliary module

At the proposal stage, we build upon the strategy in Bi et al. (2025a) to constrain 3D proposals. Unlike (Bi et al., 2025a), our method preferentially selects keypoints that are regarded as foreground points in Section 3.3 during keypoint selection. As shown in Fig. 5, spherical denoising auxiliary (SDA) denotes the denoising auxiliary strategy based on spherical neighborhoods, whereas cylindrical denoising auxiliary (CDA) denotes the cylindrical denoising auxiliary strategy adopted in this paper. Compared with spherical neighborhoods, cylindrical constraints focus more on local region modeling in the BEV plane while maintaining appropriate tolerance along the height dimension. This design better matches the spatial distribution of objects in autonomous driving scenarios, where objects are usually located near the ground plane. It helps preferentially select high score foreground points around proposal boxes and reduces the interference of background clutter during keypoint feature aggregation.

Let  $\mathbf{o}_i$  denote the center of the  $i$ th proposal box, and let  $l_i$ ,  $w_i$ , and  $h_i$  denote its length, width, and height, respectively. The radius and height of the corresponding cylindrical region are defined as:

$$r_i = \frac{1}{2} \max(l_i, w_i) + r^{(s)}, \quad e_i = h_i + r^{(s)}, \quad (14)$$

where  $r^{(s)}$  denotes the additional expansion scale. Based on the above cylindrical constraint, keypoints that are more relevant to the object can be preferentially selected near the proposal box, thereby reducing the interference of background clutter on keypoint representation.

Finally, we aggregate keypoint features at these proposal locations and further fuse the aggregated features with the BEV features obtained in Section 3.4. The fused representation is then fed into the detection head for final prediction.

### 3.6. Detection head and loss function

After obtaining the keypoint features, the detection head performs classification and regression for each proposal region, and finally outputs the object category, 3D center location, and other bounding box parameters. Let  $\mathcal{L}_{cls}$  denote the classification loss and  $\mathcal{L}_{reg}$  denote the regression loss. The total loss function is defined as:

$$\mathcal{L} = \mathcal{L}_{cls} + \mathcal{L}_{reg}, \quad (15)$$

where  $\mathcal{L}_{cls}$  is used to optimize object category prediction, and  $\mathcal{L}_{reg}$  is used to constrain the regression accuracy of 3D bounding box parameters. With the above training objective, the model can be optimized in an end to end manner from radar point cloud input to 3D bounding box prediction.

## 4. Experiments and visualization

### 4.1. Experimental setup

#### 4.1.1. Datasets

The nuScenes dataset (Caesar et al., 2020) is a widely used benchmark for autonomous driving perception with millimeter wave radar.

However, its radar point clouds lack the elevation information provided by 4D imaging radar. To evaluate the proposed method, we conduct experiments on two public 4D radar 3D object detection datasets, namely View of Delft (VoD) (Palffy et al., 2022) and TJ4DRadSet (Zheng et al., 2022). The VoD dataset contains three object categories, including car, pedestrian, and cyclist, while TJ4DRadSet contains four categories, namely car, pedestrian, cyclist, and truck. The point clouds in both datasets are mainly collected from forward facing onboard perception scenarios, covering typical traffic environments such as urban roads and open roads. The two datasets were collected in different regions with different radar configurations and exhibit clear differences in sensing range, point cloud sparsity, object categories, and scene distributions. In addition, TJ4DRadSet includes elevated roads, complex intersections, industrial areas, and scenes under strong illumination and darkness, providing more diverse conditions for evaluating the proposed method.

#### 4.1.2. Evaluation metrics

During evaluation, we follow the official settings of each dataset. The IoU threshold is set to 0.5 for car and truck, and 0.25 for pedestrian and cyclist. For the VoD dataset, two evaluation protocols are adopted. The first reports the mean average precision (mAP) over the entire annotated area, with a maximum range of 50 m. The second reports the mAP within the driving corridor area, defined as  $\{-4 \text{ m} < x < 4 \text{ m}, 0 < y < 25 \text{ m}\}$ . For the TJ4DRadSet dataset, 3D mAP and BEV mAP are evaluated within a range of 70 m from the radar sensor.

#### 4.1.3. Implementation details

The proposed model is implemented based on the OpenPCDet framework and trained on four NVIDIA GeForce RTX 3090 GPUs, each with 24 GB of memory. During voxelization, both datasets adopt a BEV grid resolution of  $0.16 \text{ m} \times 0.16 \text{ m}$ . The voxel size is set to  $[0.16, 0.16, 5]$  for VoD and  $[0.16, 0.16, 6]$  for TJ4DRadSet, with at most 16 points retained in each voxel. During keypoint sampling, the number of sampled keypoints is set to 1024.

The model is trained using the Adam One-Cycle optimization strategy. The batch size is set to 4 on each GPU, and the model is trained for 60 epochs with an initial learning rate of 0.01. During training, data augmentation strategies including random flipping along the  $x$  axis, random rotation, and random scaling are applied to improve the generalization ability of the model. In our experiments, the point cloud range is set to  $[0, -25.6, -3, 51.2, 25.6, 2]$  for VoD and  $[0, -39.68, -4, 69.12, 39.68, 2]$  for TJ4DRadSet.

### 4.2. Experiments on the VoD dataset

Table 1 reports the experimental results of the proposed method on the VoD dataset. In the modality column, L, R, and C denote LiDAR, 4D radar, and camera, respectively, while R+C denotes radar-camera fusion. It can be observed that, using only 4D radar as the single modal input, the proposed method achieves 52.27% mAP in the entire annotated area and 70.63% mAP in the driving corridor area. Notably, although no image information is introduced, the overall performance of the proposed method is comparable to that of several 4D radar camera fusion methods, and even outperforms them on some metrics. This indicates that the proposed foreground enhancement and cross branch fusion mechanisms can effectively improve the discriminative representation of object related radar features.

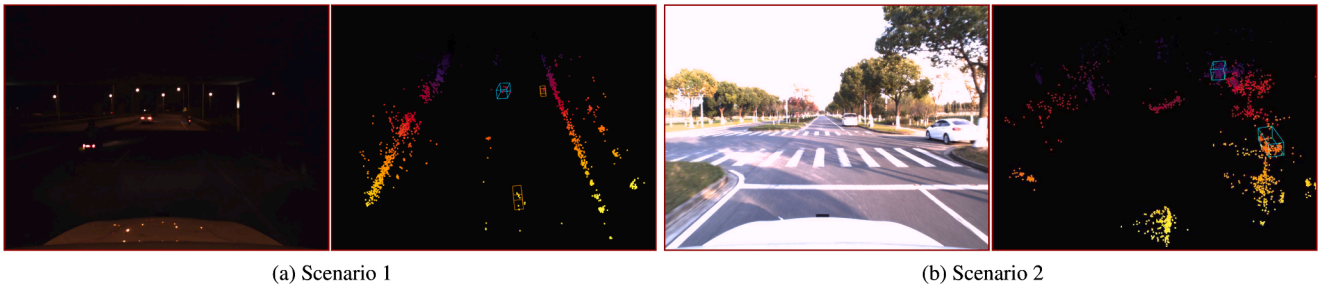
In terms of inference efficiency, the proposed method achieves 32 fps on the VoD dataset. This speed is not only higher than the data acquisition rate of common 4D radar sensors ( $\sim 15$  fps), but also satisfies the basic real-time requirement of autonomous driving perception tasks ( $\sim 10$  fps) (Bi et al., 2025a). These results demonstrate that the proposed method maintains competitive detection accuracy while achieving favorable inference speed, leading to an effective balance between accuracy and efficiency in sparse radar scenarios.

**Table 1**  
Detection performance comparison on the VoD dataset.

| Source              | Models                                  | Modality | Entire Annotated Area (%) |              |              |              | Driving Corridor Area (%) |              |              |              | FPS  |
|---------------------|-----------------------------------------|----------|---------------------------|--------------|--------------|--------------|---------------------------|--------------|--------------|--------------|------|
|                     |                                         |          | Car                       | Ped.         | Cyc.         | mAP          | Car                       | Ped.         | Cyc.         | mAP          |      |
| CVPR (2019)         | PointPillars (Lang et al., 2019)        | L→R      | 37.06                     | 35.04        | 63.44        | 45.18        | 70.15                     | 47.22        | 85.07        | 67.48        | 45.5 |
| Sensors (2018)      | SECOND (Yan et al., 2018)               | L→R      | 39.53                     | 36.23        | 65.68        | 47.14        | 71.85                     | 46.95        | 86.68        | 68.49        | 51.3 |
| CVPR (2020)         | PV-RCNN (Shi et al., 2020)              | L→R      | 40.90                     | 41.89        | <b>73.61</b> | 52.13        | 72.11                     | 51.78        | 85.89        | 69.93        | 22.6 |
| ITSC (2024)         | RadarPillarNet (Musiat et al., 2024)    | R        | 39.30                     | 35.10        | 63.63        | 46.01        | 71.65                     | 42.80        | 83.14        | 65.86        | –    |
| ICONIP (2023)       | MVFAN (Yan & Wang, 2023)                | R        | 34.05                     | 27.27        | 57.14        | 39.42        | 69.81                     | 38.65        | 84.87        | 64.38        | 45.1 |
| Appl. Sci. (2024)   | SRFF (Ruddat et al., 2024)              | R        | 36.70                     | 36.80        | 65.00        | 46.20        | 69.10                     | 47.20        | 84.30        | 66.90        | –    |
| IEEE TIV (2024)     | SMURF (Liu et al., 2023b)               | R        | 42.31                     | 39.09        | 71.50        | 50.97        | 71.74                     | 50.54        | 86.87        | 69.72        | –    |
| IEEE RAL (2026)     | RadarGaussianDet3D (Xiong et al., 2026) | R        | 40.70                     | 42.40        | 73.00        | 52.00        | 71.20                     | 51.70        | 89.00        | 70.60        | –    |
| IEEE TIM (2026)     | LITE (Yu et al., 2026)                  | R        | 40.44                     | 31.91        | 72.67        | 48.34        | –                         | –            | –            | –            | –    |
| ICRA (2023)         | BEVFusion (Liu et al., 2023c)           | R + C    | 37.85                     | 40.96        | 68.95        | 49.25        | 70.21                     | 45.86        | <b>89.48</b> | 68.52        | 7.1  |
| IEEE TIM (2023)     | RCFusion (Zheng et al., 2023)           | R + C    | 41.70                     | 38.95        | 68.31        | 49.65        | 71.87                     | 47.50        | 88.33        | 69.23        | –    |
| CVPR (2023)         | FUTR3D (Chen et al., 2023)              | R + C    | <b>46.01</b>              | 35.11        | 65.98        | 49.03        | 78.66                     | 43.10        | 86.19        | 69.32        | 7.3  |
| IEEE SENS J (2024)  | TL-4DRCF (Zhang et al., 2024)           | R + C    | 43.71                     | 40.11        | 64.22        | 49.35        | <b>79.49</b>              | 53.76        | 76.50        | 69.92        | –    |
| CVPR (2024)         | RCBEVDet (Lin et al., 2024)             | R + C    | 40.63                     | 38.86        | 70.48        | 49.99        | 72.48                     | 49.89        | 87.01        | 69.80        | –    |
| EAAI (2025)         | PSFusion (Bi et al., 2025b)             | R + C    | 42.31                     | 42.37        | 70.19        | 51.62        | 72.22                     | 50.64        | 87.81        | 70.22        | –    |
| <b>FGDBL (Ours)</b> |                                         | R        | 40.21                     | <b>44.83</b> | 71.77        | <b>52.27</b> | 69.88                     | <b>56.94</b> | 85.07        | <b>70.63</b> | 32   |

**Table 2**  
Detection performance comparison on the TJ4DRadSet dataset.

| Source              | Models                               | Modality | 3D (%)       |              |              |              |              | BEV (%)      |              |              |              |              | FPS  |
|---------------------|--------------------------------------|----------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------|
|                     |                                      |          | Car          | Ped.         | Cyc.         | Tru.         | mAP          | Car          | Ped.         | Cyc.         | Tru.         | mAP          |      |
| CVPR (2019)         | PointPillars (Lang et al., 2019)     | L→R      | 21.26        | 28.33        | 52.47        | 11.18        | 28.31        | 38.34        | 32.26        | 56.11        | 18.19        | 36.23        | 42.9 |
| CVPR (2023)         | PillarNeXt (Li et al., 2023)         | L→R      | 22.33        | 23.48        | 53.01        | 17.99        | 29.20        | 36.84        | 25.17        | 57.07        | 23.76        | 35.71        | –    |
| Sensors (2018)      | SECOND (Yan et al., 2018)            | L→R      | 18.18        | 24.43        | 32.36        | 14.76        | 22.43        | 36.02        | 28.58        | 39.75        | 19.35        | 30.93        | 24.5 |
| CVPR (2020)         | PV-RCNN (Shi et al., 2020)           | L→R      | 24.79        | 21.11        | 45.34        | 23.49        | 28.68        | 38.10        | 25.62        | 46.48        | 32.14        | 35.58        | –    |
| ITSC (2021)         | RPFA-Net (Xu et al., 2021)           | R        | 26.89        | 27.36        | 50.95        | 14.46        | 29.91        | 42.89        | 29.81        | 57.09        | 25.98        | 38.94        | –    |
| ITSC (2024)         | RadarPillarNet (Musiat et al., 2024) | R        | 28.45        | 26.24        | 51.57        | 15.20        | 30.37        | 45.70        | 29.19        | 56.89        | 25.17        | 39.24        | –    |
| ICANN (2024)        | MUFASA (Peng et al., 2024)           | R        | 23.56        | 23.70        | 48.39        | 25.25        | 30.23        | 41.25        | 24.54        | 53.64        | <b>36.97</b> | 39.10        | –    |
| IEEE TIM (2026)     | LITE (Yu et al., 2026)               | R        | 18.92        | <b>31.98</b> | <b>60.12</b> | 20.31        | 32.83        | 30.44        | <b>35.83</b> | <b>62.85</b> | 31.37        | 40.12        | –    |
| ICRA (2019)         | MVX-Net (Sindagi et al., 2019)       | R + C    | 22.28        | 19.57        | 50.70        | 11.21        | 25.94        | 37.46        | 22.70        | 54.69        | 18.07        | 33.23        | –    |
| IEEE TIM (2023)     | RCFusion (Zheng et al., 2023)        | R + C    | 29.72        | 27.17        | 54.93        | 23.56        | 33.85        | 40.89        | 30.95        | 58.30        | 28.92        | 39.76        | 4.7  |
| CVPR (2024)         | RCBEVDet (Lin et al., 2024)          | R + C    | 29.38        | 28.13        | 49.60        | 17.37        | 31.12        | 41.77        | 28.98        | 53.03        | 26.05        | 37.46        | –    |
| ICCV (2025)         | CVFusion (Zhong et al., 2025)        | R + C    | <b>54.08</b> | 26.90        | 40.88        | 22.16        | <b>36.00</b> | <b>60.22</b> | 28.17        | 42.82        | 30.96        | 40.54        | 8.7  |
| <b>FGDBL (Ours)</b> |                                      | R        | 26.41        | 31.38        | 54.67        | <b>25.26</b> | 34.43        | 37.49        | 33.78        | 56.65        | 34.56        | <b>40.62</b> | 23   |



**Fig. 6.** Visualization results of 4D radar 3D detection on TJ4DRadSet.

#### 4.3. Experiments on the TJ4DRadSet dataset

Furthermore, Table 2 presents the experimental results of the proposed method on the TJ4DRadSet dataset. Compared with VoD, TJ4D RadSet contains more long range objects and open road scenarios, with a larger perception range and sparser point clouds. Therefore, it imposes higher requirements on the robustness and generalization ability of detection models. As shown in Table 2, the proposed method achieves 34.43% 3D mAP and 40.62% BEV mAP, showing competitive performance compared with various state of the art methods. This

demonstrates that the proposed method still maintains good generalization ability in challenging scenarios such as long range perception and open road environments.

Meanwhile, Fig. 6 shows the visualization results of the proposed model on TJ4DRadSet scenes. It can be clearly observed that 4D radar point clouds are highly sparse: a single object is usually represented by only a few points, while a large number of isolated noise points and background clutter exist in the scene. Under such conditions, without explicit foreground background distinction, the detection model is easily disturbed by clutter, causing object responses to be weakened or even

**Table 3**  
Experiments with different numbers of keypoints.

| Points | 3D (%) |       |       |       |       | BEV (%) |       |       |       |       |
|--------|--------|-------|-------|-------|-------|---------|-------|-------|-------|-------|
|        | Car    | Ped.  | Cyc.  | Tru.  | mAP   | Car     | Ped.  | Cyc.  | Tru.  | mAP   |
| 128    | 24.35  | 32.73 | 50.07 | 24.96 | 33.03 | 35.97   | 33.82 | 53.11 | 33.98 | 39.22 |
| 256    | 25.25  | 31.81 | 52.93 | 23.64 | 33.41 | 36.90   | 32.88 | 54.68 | 34.11 | 39.64 |
| 512    | 28.48  | 30.00 | 53.02 | 25.72 | 34.31 | 38.78   | 31.09 | 57.84 | 32.81 | 40.13 |
| 1024   | 26.41  | 31.38 | 54.67 | 25.26 | 34.43 | 37.49   | 33.78 | 56.65 | 34.56 | 40.62 |
| 2048   | 27.26  | 31.29 | 54.91 | 24.97 | 34.61 | 35.96   | 34.80 | 55.44 | 33.89 | 40.02 |

**Table 4**  
Ablation study of the proposed modules.

| Method |        |     | 3D (%) |       |       |       |              | BEV (%) |       |       |       |              |
|--------|--------|-----|--------|-------|-------|-------|--------------|---------|-------|-------|-------|--------------|
| DLQG   | FGGCBF | CDA | Car    | Ped.  | Cyc.  | Tru.  | mAP          | Car     | Ped.  | Cyc.  | Tru.  | mAP          |
| –      | –      | –   | 24.79  | 21.11 | 45.34 | 23.49 | 28.68        | 38.10   | 25.62 | 46.48 | 32.14 | 35.58        |
| ✓      | –      | –   | 26.33  | 33.30 | 48.69 | 20.18 | 32.13        | 35.97   | 34.87 | 49.74 | 26.36 | 36.74        |
| –      | ✓      | –   | 26.01  | 24.76 | 49.93 | 22.01 | 30.68        | 37.32   | 32.60 | 56.12 | 25.15 | 37.79        |
| –      | –      | ✓   | 24.84  | 30.64 | 47.87 | 19.46 | 30.70        | 30.66   | 33.39 | 55.52 | 24.03 | 35.90        |
| –      | ✓      | ✓   | 27.31  | 27.54 | 49.52 | 26.92 | 32.82        | 39.63   | 32.17 | 50.85 | 30.62 | 38.32        |
| ✓      | –      | ✓   | 28.84  | 30.18 | 51.20 | 23.49 | 33.43        | 37.52   | 33.71 | 54.40 | 34.10 | 39.93        |
| ✓      | ✓      | –   | 25.33  | 29.72 | 53.62 | 26.33 | 33.75        | 35.54   | 31.94 | 54.73 | 33.67 | 38.97        |
| ✓      | ✓      | ✓   | 26.41  | 31.38 | 54.67 | 25.26 | <b>34.43</b> | 37.49   | 33.78 | 56.65 | 34.56 | <b>40.62</b> |

overwhelmed. Through foreground guided query generation and gated fusion, the proposed method can strengthen effective object related responses in sparse point clouds and suppress invalid interference from background regions. As a result, the detection stability and robustness in complex scenarios are improved. This observation is further supported by the following experiments. Considering the complexity of the TJ4D RadSet dataset, the following experiments are mainly conducted on this dataset.

#### 4.4. Effect of the number of keypoints

To analyze the influence of the number of sampled keypoints on detection performance, we conduct comparative experiments with different numbers of keypoints on the TJ4DRadSet dataset. The results are reported in Table 3. It can be observed that when the number of keypoints is too small, the model fails to preserve sufficient object structural information, which clearly limits detection performance. As the number of keypoints increases, both 3D detection performance and BEV detection performance show an overall upward trend. When the number of keypoints increases from 128 to 1024, the 3D mAP improves from 33.03% to 34.43%, and the BEV mAP increases from 39.22% to 40.62%. This indicates that appropriately increasing the number of keypoints helps preserve more complete object information, thereby improving detection performance.

Furthermore, when the number of keypoints continues to increase to 2048, the 3D mAP slightly improves to 34.61%, but the BEV mAP decreases to 40.02%, resulting in limited overall gains. This is because excessive keypoints may introduce additional background clutter and noisy points while bringing more local details. These irrelevant points may weaken the discriminative capability of keypoint representations and increase computational cost. Considering both detection accuracy and efficiency, we set the number of keypoints to 1024 as the default configuration for subsequent experiments. This setting achieves a favorable balance between information preservation and noise suppression.

A core innovation of this study is to estimate point-wise foreground importance through a scoring mechanism. To verify the intermediate process of the proposed module, we visualize the foreground point selection process, as shown in Fig. 7. It should be noted that radar points are projected onto the image plane only for visualization, while the actual experiments still use 4D radar as the single modal input. From Fig. 7(a), it can be observed that the raw point cloud contains many isolated noise points in background regions such as trees and buildings,

whereas real objects are usually represented by only a few radar points. Fig. 7(b) shows the importance estimation results for all radar points, where different colors represent different point-wise scores. It can be seen that object related points usually obtain higher scores, while points in background clutter regions show weaker responses. Fig. 7(c) further presents the foreground points retained after foreground guided selection. The retained points are mainly concentrated around object regions such as vehicles, pedestrians, and cyclists, while a large number of background clutter points are effectively suppressed. These results indicate that the proposed foreground point scoring and gating mechanism can effectively distinguish foreground responses from background clutter, providing more reliable foreground cues for subsequent feature fusion and proposal refinement.

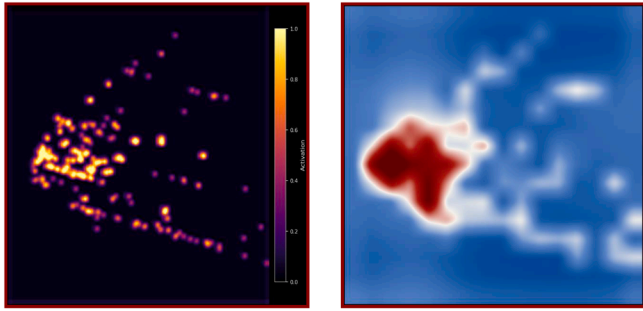
#### 4.5. Ablation study

To verify the effectiveness of each component, we conduct an ablation study on the proposed modules and compare the results with the baseline model PV-RCNN (Shi et al., 2020). All the proposed modules are implemented on a PV-RCNN-based detection framework, and sparsity pillar attention (Bi et al., 2025a) is adopted to facilitate global feature interaction. All ablation configurations use the same training settings for a fair comparison. The results are shown in Table 4. Considering that the proposed modules are functionally coupled, for example, DLQG generates foreground cues and FGGCBF further uses these cues to guide feature interaction, we analyze both the individual contribution of each module and the effects of different module combinations. For the baseline model without DLQG, FGGCBF, and CDA, the 3D mAP and BEV mAP are 28.68% and 35.58%, respectively. The individual module results show that each component improves the detection performance to varying degrees, while different module combinations provide further gains. When all three modules are enabled simultaneously, the model achieves the best performance, with 34.43% 3D mAP and 40.62% BEV mAP. Compared with the baseline, the improvements are 5.75 and 5.04 percentage points, respectively, demonstrating that the proposed modules provide clear synergistic gains in sparse 4D radar scenarios.

To further analyze the role of the auxiliary branch, Fig. 8 visualizes the auxiliary branch feature activation map and the cross attention response map. Specifically, Fig. 8(a) shows the feature activation distribution of the auxiliary branch, while Fig. 8(b) presents the cross attention token importance map during the fusion stage. It can be observed that the high response regions in the auxiliary branch show clear



Fig. 7. Visualization of foreground point scoring and selection process.



(a) Auxiliary feature activation map (b) Cross-attention token importance map

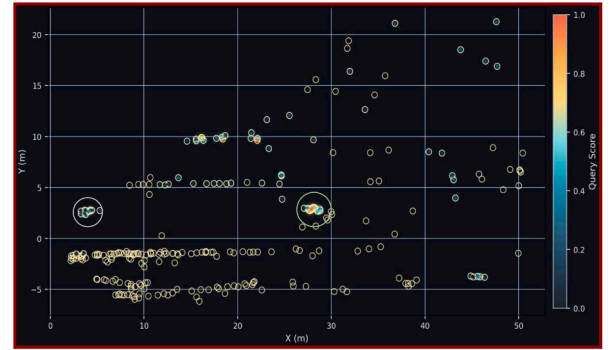
Fig. 8. Visualization of auxiliary branch activation and cross attention response.

spatial correspondence with the high weight regions in cross attention, and their distributions exhibit strong consistency. This indicates that the foreground cues extracted by the auxiliary branch can effectively guide cross branch attention to focus on potential object regions rather than being dominated by extensive background clutter. In other words, the auxiliary branch is not simply an additional branch. Instead, it provides discriminative foreground guidance for the main branch in sparse and cluttered 4D radar scenarios, thereby validating the effectiveness of the proposed foreground guided fusion strategy.

#### 4.6. Comparison of denoising strategies

Finally, we evaluate different denoising strategies, and the results are shown in Table 5. It can be observed that, without introducing any denoising strategy, the model achieves 33.75% 3D mAP and 38.97% BEV mAP. After adopting SDA, the performance is slightly improved. When CDA is used, the model achieves the best results, with the 3D mAP and BEV mAP increasing to 34.43% and 40.62%, respectively. These results indicate that introducing local geometric constraints at the proposal refinement stage can effectively reduce the interference of noisy points and irrelevant background points on keypoints. In particular, the cylindrical constraint shows better adaptability than the spherical constraint.

The reason is that objects in 4D radar scenarios usually exhibit more evident lateral distribution patterns in the BEV space. The cylindrical region achieves a more reasonable balance between horizontal range constraints and vertical tolerance. Therefore, it is more beneficial for preserving effective object related points while suppressing isolated noisy



(a) Query scores in the BEV view



(b) Feature responses after aggregation

Fig. 9. Visualization of foreground responses before and after CDA aggregation.

points and background clutter around proposal boxes. In contrast, although the spherical constraint can also perform local filtering to some extent, its spatial coverage is more likely to include additional irrelevant points, thereby weakening the discriminative capability of keypoint features.

Fig. 9 shows the foreground response distribution from the BEV perspective and the feature response after CDA aggregation. As shown in Fig. 9(a), high score responses are mainly concentrated around potential objects, while the responses in large background regions are relatively weak. After denoising and feature aggregation through the CDA module, as shown in Fig. 9(b), the model preserves feature responses in key

**Table 5**  
Experiments with different denoising strategies.

| Denoising | 3D (%) |       |       |       |       | BEV (%) |       |       |       |       |
|-----------|--------|-------|-------|-------|-------|---------|-------|-------|-------|-------|
|           | Car    | Ped.  | Cyc.  | Tru.  | mAP   | Car     | Ped.  | Cyc.  | Tru.  | mAP   |
| –         | 25.33  | 29.72 | 53.62 | 26.33 | 33.75 | 35.54   | 31.94 | 54.73 | 33.67 | 38.97 |
| SDA       | 28.26  | 30.87 | 52.02 | 24.61 | 33.94 | 36.10   | 31.33 | 53.71 | 35.09 | 39.06 |
| CDA       | 26.41  | 31.38 | 54.67 | 25.26 | 34.43 | 37.49   | 33.78 | 56.65 | 34.56 | 40.62 |

**Table 6**  
Robustness comparison under different point cloud retention ratios.

| Retention ratio | Ours       |           |             |            | Baseline   |           |             |            |
|-----------------|------------|-----------|-------------|------------|------------|-----------|-------------|------------|
|                 | 3D mAP (%) | 3D drop ↓ | BEV mAP (%) | BEV drop ↓ | 3D mAP (%) | 3D drop ↓ | BEV mAP (%) | BEV drop ↓ |
| 100%            | 34.43      | 0.00      | 40.62       | 0.00       | 29.55      | 0.00      | 35.73       | 0.00       |
| 90%             | 33.91      | 0.52      | 40.13       | 0.49       | 28.19      | 1.36      | 34.87       | 0.86       |
| 80%             | 33.29      | 1.14      | 39.76       | 0.86       | 27.73      | 1.82      | 34.12       | 1.61       |
| 70%             | 32.97      | 1.46      | 39.08       | 1.54       | 26.26      | 3.29      | 33.04       | 2.69       |
| 50%             | 28.51      | 5.92      | 35.51       | 5.11       | 22.69      | 6.86      | 28.86       | 6.87       |

regions and significantly suppresses background clutter responses. This phenomenon indicates that CDA not only provides more effective local geometric constraints at the proposal level, but also works synergistically with the foreground guided mechanism. As a result, the subsequent feature representation becomes more focused on real object regions, thereby improving detection performance.

#### 4.7. Robustness to point cloud degradation and distance-wise analysis

To further analyze the detection performance of the proposed method under different point cloud sparsity levels and sensing distances, we conduct point cloud degradation robustness and distance-wise evaluation on the TJ4DRadSet dataset.

In the point cloud degradation experiment, we randomly retain 100%, 90%, 80%, 70%, and 50% of the original radar points during evaluation, and compare the proposed method with the baseline model. For this robustness analysis, the baseline model was independently re-trained under the same training protocol. The results are reported in Table 6.

As the point cloud retention ratio decreases, the detection performance of both models declines. However, the proposed method exhibits a slower performance degradation. When 70% of the radar points are retained, the 3D mAP and BEV mAP of our method decrease by only 1.46 and 1.54 percentage points, respectively, whereas those of the baseline decrease by 3.29 and 2.69 percentage points. Even under the severe degradation condition in which only 50% of the radar points are retained, our method still achieves 28.51% 3D mAP and 35.51% BEV mAP, clearly outperforming the baseline under the same condition. Considering that TJ4DRadSet already contains long range objects, sparse radar returns, and complex background interference, randomly removing half of the radar points constitutes a relatively severe stress test. These results demonstrate that the proposed method maintains favorable robustness under substantial point cloud degradation.

In the distance-wise experiment, we divide the evaluation range into near, middle, and far range intervals according to the horizontal distance between the object center and the radar sensor. The results are presented in Table 7.

The model achieves 52.77% 3D mAP and 59.88% BEV mAP within the range of 0–25 m, demonstrating strong near-range detection capability. As the object distance increases, the number of radar returns and the available geometric information gradually decrease, resulting in an evident performance decline. In particular, the detection performance remains limited within the far range interval of 50–70 m. The limited far range performance is mainly associated with the extremely sparse returns of distant small objects, particularly pedestrians and cyclists, which may produce only a few or even a single valid radar return. Al-

**Table 7**  
Detection performance across different distance ranges.

| Distance group | Range (m)    | 3D mAP (%) | BEV mAP (%) |
|----------------|--------------|------------|-------------|
| All            | [0.0, 70.0)  | 34.43      | 40.62       |
| Near           | [0.0, 25.0)  | 52.77      | 59.88       |
| Middle         | [25.0, 50.0) | 18.81      | 26.29       |
| Far            | [50.0, 70.0) | 6.13       | 6.37        |

though far range objects may not immediately trigger emergency collision avoidance, their reliable perception remains important for anticipatory planning and safe decision making. Therefore, these results reflect the inherent difficulty of detecting distant small objects and also indicate a limitation that should be further addressed in future work.

## 5. Conclusion

This paper addresses the problem that object responses in sparse 4D radar point clouds are easily disturbed by background clutter, and proposes a foreground guided dual branch learning framework, termed FGDBL. The proposed framework generates point-wise foreground cues through DLQG and selectively injects auxiliary foreground information into the main branch features through FGGBCF. Meanwhile, CDA further constrains keypoint selection at the proposal stage. In this way, a relatively complete foreground enhancement pipeline is formed, covering point-wise scoring, feature fusion, and local refinement. Experimental results demonstrate that the proposed method improves object representation in sparse radar scenarios and achieves competitive detection performance under real-time inference.

Although the proposed method achieves promising performance using only 4D radar input, the single radar modality still has certain limitations in fine-grained semantic representation and complex occlusion understanding. Moreover, the foreground scoring mechanism relies on local density and motion consistency, which may limit its effectiveness for extremely sparse or single-return objects. The current framework does not explicitly model more complex radar physical and temporal priors, and its evaluation is limited to two public datasets and desktop GPU platforms. Therefore, further evaluation is still required regarding its broader interpretability, cross-dataset generalization, embedded deployment, and applicability to production-level safety validation and failure traceability. In future work, we will further explore multimodal fusion between 4D radar and camera image features, introduce richer visual semantic priors, and investigate more robust foreground guided fusion mechanisms under sensor degradation, calibration errors, and adverse weather conditions.

## CRedit authorship contribution statement

**Yuanhang Wang:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Yonghua Zhou:** Writing – review & editing, Supervision, Resources, Investigation, Funding acquisition; **Yongnan Zhang:** Writing – review & editing, Funding acquisition, Conceptualization; **Yiduo Mei:** Writing – review & editing, Supervision, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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